Md. Danial Dirar

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ML and Software Engineer building local-first AI systems, generative media pipelines, and forecasting models

Experience

Business Development Executive — *Dhaka, Bangladesh*
Codexa IT LLC — *August 2026 - Present*

- Support business development for a US-based property preservation outsourcing company, working with its American clients and field vendor network.

Founder — *Dhaka, Bangladesh*
Haifa Intelligence, AI/ML Studio — *July 2026 - Present*

- Run an independent studio building local-first, privacy-respecting AI tools (haifa-ai.com); shipped Haifa HiveMind, a local AI research assistant that runs on the user's own hardware with no cloud dependency.
- Build generative image and video pipelines in ComfyUI with custom LoRA models for advertising creatives, running entirely on local GPU hardware.

Founder and Owner — *Dhaka, Bangladesh*
Steam Store Bangladesh (SSB) — *July 2016 - 2026*

- Founded and ran an online store selling digital products for ten years, handling day-to-day operations, PR, and customer communication.

Chief Communication Officer — *Dhaka, Bangladesh*
AFK Productions — *February 2025 - June 2025*

- Led the communications department as Head of Communications for esports events run with brands including MSI, ASUS ROG, and Acer Predator.

Entry Level Software Engineer (Part-Time) — *Dhaka, Bangladesh*
Arabian Gulf Associates Co. Ltd — *July 2024 - October 2024*

- Worked part-time as a technical IT engineer in a recruiting agency.

Education

BRAC University — *June 2022 - Present*
B.Sc. (Hons.) in Computer Science and Engineering

- **Thesis:** Forecasting groundwater levels in the Barind region of northwest Bangladesh from historical hydrological and environmental data, using PyTorch models that currently reach an R^2 of about 0.93.

Publications

Design and field performance of an integrated horizontal flow treatment managed aquifer recharge system for groundwater rehabilitation in Bangladesh — Oct 2025
Hossain MI, Bari MN, **Dirar MD**, et al. *Applied Water Science* 15:272, Springer Nature. DOI

Engineering hydrological resilience: Quantitative assessment of managed aquifer recharge (MAR) potential utilizing advanced horizontal flow treatment in water stressed regions — Dec 2025
Hossain MI, Bari MN, **Dirar MD**, et al. *Applied Water Science* 16:7, Springer Nature. DOI

Technologies

Core Expertise: Machine Learning, Deep Learning, Time-Series Forecasting, Generative AI, Local AI, RAG Systems, Business Tailored Models

Languages: Python, C, C++, C#, TypeScript, Dart, Lua

ML & Data: PyTorch, LightGBM, Optuna, SHAP, Polars, CLIP, DINOv2

Generative AI: ComfyUI, custom LoRA, LTX-2.5, Wan 2.5, Qwen3-VL, Krea2, MinimaxH3

Web & Mobile: Next.js, NestJS, Node.js, Flutter, shadcn/ui, PostgreSQL, Firebase, Supabase

Systems & Tools: Linux (Arch-based), Git, OpenGL, ArcGIS, QGIS

Projects

FictiPay Churn Prediction (bKash x NSUCEC Datathon) *Polars, LightGBM, Optuna, SHAP*

- Reached a leaderboard AUC of 0.98451 with Optuna-tuned LightGBM and Logistic Regression models.
- Used SHAP to verify zero temporal leakage and confirmed a hard signal ceiling in the data near 0.9844.

Generative Image and Video Pipelines *ComfyUI, RTX 3090*

- Built local text-to-video and character replacement workflows on LTX-2.3 (22B) and Wan 2.1, documenting valid resolutions, frame counts, and CFG behaviour for repeatable output.
- Built a Krea2 Turbo generate-then-upscale pipeline and a Qwen3-VL image-to-prompt generator.

Multi-Modal Vision Meta-Learning Framework *CLIP, DINOv2, FOMAML*

- Built a vision meta-learning framework on CLIP and DINOv2 backbones with first-order MAML (FOMAML).

CreditSense AI *Machine Learning*

- Built a loan eligibility prediction pipeline designed to be leak-free from preprocessing through evaluation.

Resource-Aware Distributed ML Training *Distributed Systems*

- Built a master-worker system that trains regression models across nodes, balancing work by node resources.

Metagenomic Profiling of Plastic-Degrading Microbial Communities *DIAMOND, BLAST*

- Profiled plastic-degrading microbial communities in urban water bodies from NIH gene expression data.

Global River Database *TypeScript, Node.js, PostgreSQL*

- Designed and implemented a global river database with a hierarchy-based verification method for data integrity.

Signal Sweep *Computer Graphics*

- Built a 3D radar interface that scans for nearby Bluetooth devices and plots them with interactive controls.

Haifa OS *Linux, Bootloader*

- Built a Linux distribution with a modified bootloader, aimed at a smooth experience for new users of all kinds.

DishDash *Android, iOS*

- Developed a cross-platform mobile app for discovering nearby restaurant offers using real-time location.

Other Work

- Alzheimer's disease prediction, obesity risk prediction, pet segmentation and breed classification, e-commerce shipping optimization, a Dijkstra vs A* grid path planning benchmark and some website projects for clients.

Achievements

6th Place, National Programming Contest 2017

Placed sixth nationally in a programming contest organised by a2i, Bangladesh.

Languages

Bengali (Native), English (Bilingual proficiency), Spanish (Basic), Japanese (Basic)